Part I: User Research & Data Collection

Introduction

The goal of my project is to assess the ease with which users can examine and remediate accessibility issues in their Google Docs documents. Currently, Google Docs lacks a fully integrated accessibility tool, requiring users to download third-party add-ons such as Grackle to identify accessibility problems. However, Grackle presents its own challenges: it has a cluttered interface with confusing features, operates slowly, and lacks user-friendliness. These limitations make it difficult for content creators and reviewers to efficiently navigate and meet accessibility criteria.

This problem is particularly urgent given my work context. Our team supports professors in complying with the new ADA Title II digital accessibility law, with a compliance deadline of April 2026 rapidly approaching. We process high volumes of documents weekly for numerous clients, working to make materials accessible while also providing resources and guidance that enable clients to remediate materials independently. For this self-service approach to succeed, the tools and resources we recommend must be intuitive and accessible to users with varying levels of technical expertise. The current limitations of available tools directly impact our ability to help clients meet their legal obligations efficiently.

My initial design goal was to identify users' primary pain points and incorporate their feedback into an integrated accessibility platform, like the accessibility checker built into Microsoft PowerPoint.

I selected two participants to provide contrasting perspectives. My first participant is a coworker who regularly uses Grackle in our day-to-day work and has extensive familiarity with document accessibility standards. My second participant is a third-year UCI student with no prior experience using Grackle and limited knowledge of digital accessibility. This pairing allows me to capture both expert insights on recurring usability issues and a fresh user's perspective focused primarily on UX/UI elements and initial impressions.

**It's important to note that my focus extends beyond critiquing Grackle specifically. I aim to evaluate the entire process of remediating document accessibility. However, since Grackle is currently the primary tool available for this purpose in Google Docs, it will feature prominently throughout my observation and project analysis.

Observation Plan

Each observation session will last approximately 15 minutes. During this time, participants will work on improving document accessibility using Grackle Docs. I will conduct the first observation in-person with my coworker, while the second observation will be conducted remotely via FaceTime with screen sharing.

Participants will use Grackle to guide their accessibility remediation process. For those who do not have Grackle installed, I will provide brief setup instructions before beginning the observation.

Follow-up Questions:

1. Do the accessibility issues identified by Grackle make sense to you?
2. Would you describe this extension as user-friendly?
3. What aspects of the process did you find most helpful?
4. What was the most frustrating element you encountered while using Grackle?
5. Does having this tool make reviewing document accessibility easier compared to not having any guidance at all?
6. If you could change one thing about this process, what would it be?

Challenges the Participant Encountered

Participant 1:

Issue 1: Due to significant lag, the participant often clicked multiple times on an issue, assuming her initial click hadn't registered, when in reality the system was simply slow to respond. These repeated clicks caused the issue resolution pop-up to glitch and attempt to open multiple instances simultaneously, creating frustration.

Issue 2: Fixing certain issues required a two-step process: first pressing "Locate," then pressing "Tag." She found this unnecessarily tedious and expressed a desire for a single-button solution that would simultaneously navigate to the issue and enable immediate remediation

Participant 2:

Issue 1: The participant found the issue descriptions difficult to understand. While she acknowledged she could research the terminology to better comprehend the problems,

doing so would be time-consuming and defeat the purpose of using an accessibility tool designed to streamline the process.

Issue 2: When attempting to review all identified issues, the participant found the excessive scrolling required to reach the bottom of the list frustrating. She wished for a collapse-all function to condense the view.

Participant Reflection (Follow-Up Questions)

Participant 1:

1. **Would you say that the issues Grackle listed make sense? "***No, not all the listings make a lot of sense."*

2. **Is this extension user-friendly for you? "**It does what it needs to do but it's not the easiest to use."

3. **What did you find helpful during this process?** "*I like how it allows you to tag all the tables at once, I think it should do that for some of the other issues"*

4. **What is one frustrating thing you encountered while using Grackle?** "*The constant message to basically upgrade my plan whenever I tried to select an issue was really annoying. I'm still able to use the free version to see what the issues are and having the constant pop-up message makes it difficult to even use the free version. They should have a "do not show again" button."*

5. **Does this make reviewing the accessibility of the document easier for you [as opposed to not having it at all]?** "*It is a helpful guideline but it's definitely not the best version, there are other accessibility checkers for different platforms that are way better so I wish GDocs had one like that."*

6. **What would you change about this process? "***I wish it were easier to read and navigate this tool while checking accessibility. For example I would like the document structure section to be indented like it is in the actual document layout through Gdocs so that I could skim quicker for issues."*

Participant 2:

1. **Would you say that the issues Grackle listed make sense?** "*No, I don't get what half of this is saying and why it's an issue"*

2. Is this extension user-friendly for you? “No, I don't really like how it looks and it is super laggy.”

3. **What did you find helpful during this process?** “*It lists a lot of issues which is a helpful starting point.”*

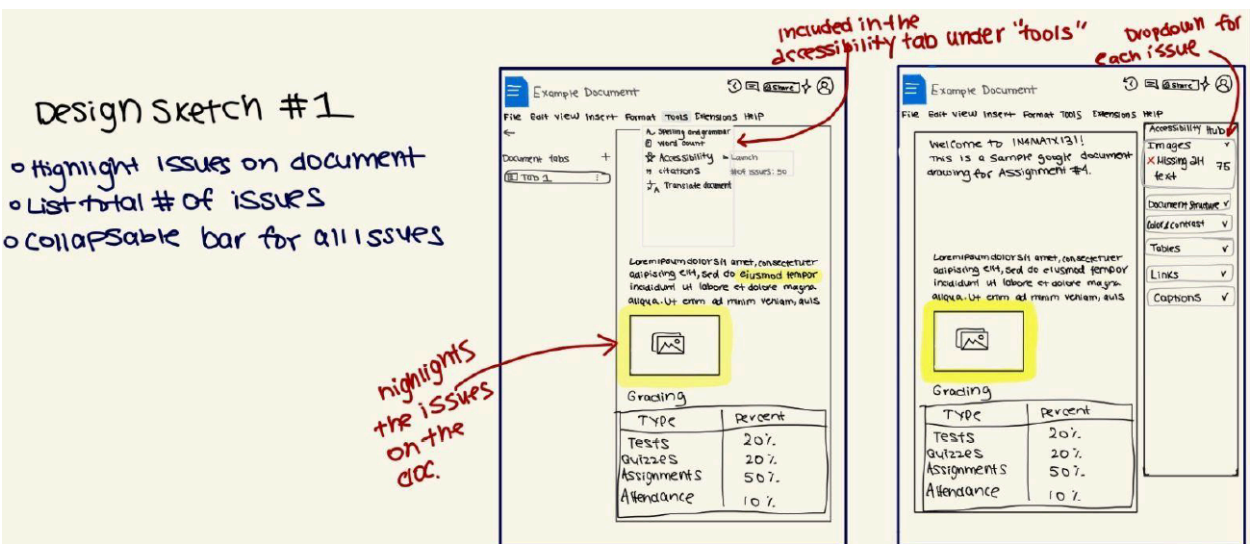
4. **What is one frustrating thing you encountered while using Grackle?** “*The lag really annoyed me and made it a frustrating experience.”*

5. **Does this make reviewing the accessibility of the document easier for you [as opposed to not having it at all]?** “*I don’t like it, I'd rather not use it.”*

6. **What would you change about this process?** “*I think each issue should be collapsible so that you aren’t scrolling for such a long time just to skim through and see all the issues.”*

Part II: Sketching

Design Sketch #1

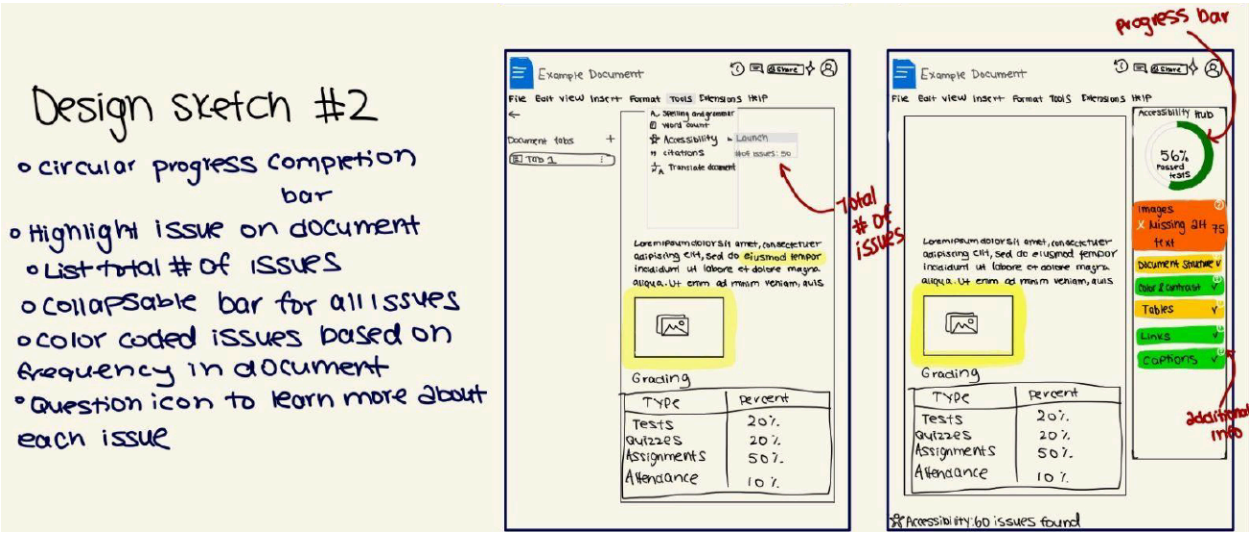


In this sketch, I've integrated an advanced accessibility feature directly into the "Tools" menu. The feature consists of two components: a quick-view summary displaying the total number of accessibility issues in the document, and an "Accessibility Hub" that provides an in-depth breakdown and streamlined navigation for locating and fixing each issue.

When users launch the Accessibility Hub, a sidebar opens displaying issues organized by category, with expandable dropdowns allowing users to prioritize which issues to address first. Upon initial opening, all accessibility issues are highlighted throughout the document (where visually applicable). Once a user selects a specific category, only the relevant issues for that category remain highlighted. Users can then drill down further by clicking on individual issues within each category—for example, "Missing alt text" under the Images category—to address them directly.

Since this solution is fully integrated into Google Docs, it eliminates the need to install third-party extensions from the extensions marketplace. The system provides live activity tracking, meaning issue counts and status updates refresh automatically as users make changes. This removes the need for constant manual re-scanning—a significant pain point identified by one of my participants.

Design Sketch #2

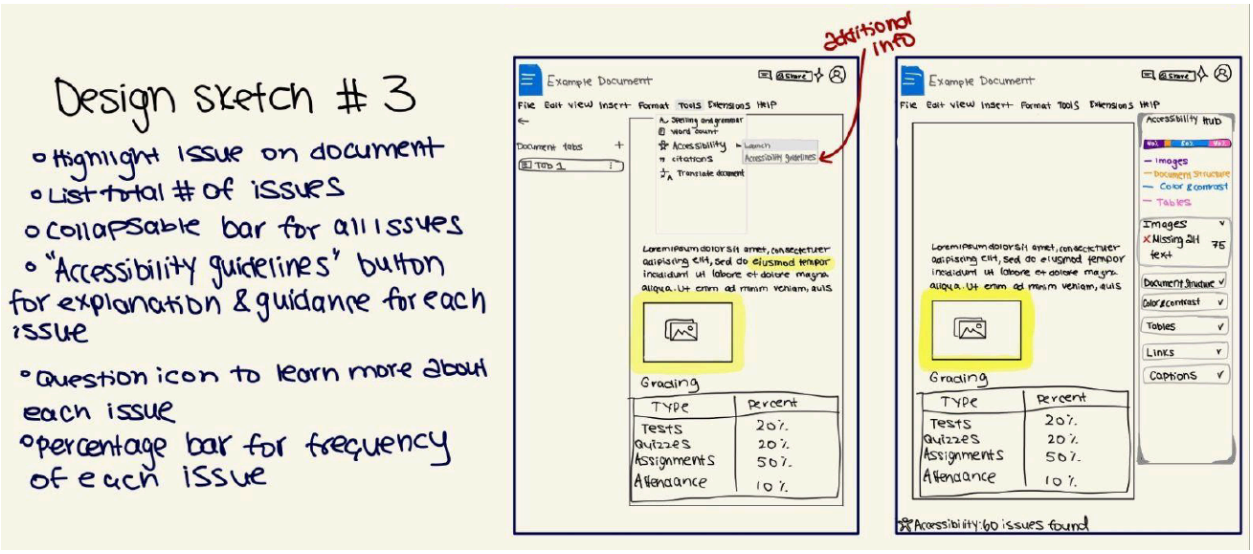


In this sketch, launching the "Accessibility Hub" displays a progress bar showing the percentage of the document that meets accessibility standards. Essentially, the proportion of accessibility tests passed. To provide additional visual context, each issue category features color-coded indicators ranging from green to red based on issue frequency: categories with numerous violations appear in red, while categories with fewer issues display in green.

Each issue category includes a question mark icon that users can click to access a concise explanation covering what the issue means, why it matters for accessibility, and best practices for remediation. This feature directly addresses feedback from one of my participants, who expressed a need for clearer, more informative guidance which is particularly valuable for users less familiar with digital accessibility standards.

I've also incorporated a persistent live status indicator at the bottom of the document, similar to the existing word count display option. This feature shows the current number of accessibility issues in real-time, eliminating the need for constant manual rescanning and providing immediate confirmation when changes successfully resolve issues. Users can toggle this display on or off based on their preference.

Design Sketch #3



This sketch incorporates most features from Designs 1 and 2, but replaces the progress bar and color-coded issue tabs with a frequency bar chart. Each issue category is assigned a distinct color, allowing users to quickly assess which areas require the most attention and where to prioritize their remediation efforts.

Rather than placing question mark icons next to individual issues, I've created a separate "Documentation" tab accessible from the main launch button. This opens a dedicated page containing comprehensive explanations for all accessibility issues in one centralized location. I believe this approach is more efficient, particularly since users already have the option to display live issue counts while editing, reducing the need for inline help elements that could clutter the interface.

Across all three designs, I prioritized three key principles: showing progress, providing clarity, and highlighting issue instances. These directly address the frustrations voiced by my participants, and my redesigns specifically target improvements in these areas.

All three layouts draw significant inspiration from the accessibility checker feature currently available in Microsoft Word and PowerPoint, which I found to be a strong model for integrated accessibility tools.

Part III: Prototyping

I decided to move forward with implement design #2

Figma Prototype Link

With the design that I created, users will be able to:

● **Monitor accessibility progress** through a percentage indicator showing how much of their document meets accessibility standards

● **Display live issue counts** at the bottom of the screen while editing, providing real-time feedback without interrupting workflow

● **Collapse and expand issue categories** for streamlined navigation and reduced scrolling

● **Visualize issues instantly** as the Accessibility Hub highlights all problems or filters to show only selected categories

● **Remediate issues directly** within the interface without navigating away from the document

● **Interpret issue severity at a glance** through color-coded indicators based on frequency: red for high occurrence, yellow for medium, and green for low or zero instances

● **Identify and fix a comprehensive range of issues**, including raw URLs and non-descriptive links, with direct remediation options

● **Utilize auto-fix functionality** for individual categories or all issues simultaneously, with the ability to review, approve, or edit automated fixes before finalizing them—a feature inspired by positive feedback from one of my participants regarding the existing system

● **Access educational resources** explaining what each issue is, why it matters for accessibility, and best practices for resolution